



Lehrstuhl für Energiesysteme

Title of Semesterarbeit: ANALYSIS AND MACHINE LEARNING MODELING
OF PLASTIC PYROLYSIS DATA

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Assigned: 25.10.2024

Handed in: 14.04.2026

Abstract

Accurate prediction of plastic pyrolysis product yields is essential for process design and optimization, yet existing machine learning studies in this domain suffer from unrecognized methodological shortcomings. This work presents a systematic investigation into machine learning-based yield prediction for non-catalytic plastic pyrolysis, built upon a unified database of 1,072 experimental data points compiled from 127 peer-reviewed publications. After systematic preprocessing - including mass balance validation, and feature harmonization - 795 data points remained for model training and evaluation.

A central contribution of this thesis is the identification and quantification of data leakage in pyrolysis machine learning. This work demonstrates that conventional random cross-validation inflates model accuracy by approximately 50 % to over 100 % in terms of mean absolute error (MAE) because data points from the same publication share latent correlations. To address this, Paper-Level GroupKFold cross-validation is introduced as the first application of grouped splitting by publication source in the pyrolysis ML literature—ensuring that no paper contributes data to both training and test folds simultaneously.

Under this rigorous evaluation protocol, the best-performing gradient-boosted models achieve MAE values of 16,41 wt% (oil), 11,97 wt% (gas), and 11,08 wt% (char) for XGBoost, which was selected as the primary model for its exact SHAP implementation and literature comparability. SHAP (SHapley Additive exPlanations) analysis confirms that the dominant feature - pyrolysis temperature - exhibits physically consistent directional effects across all three product fractions, validating the model's reliance on genuine thermochemical relationships rather than statistical artifacts. The residual prediction error is shown to approach the inter-laboratory noise floor inherent in heterogeneous literature data.



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Abbreviations

Abbreviation	Meaning
ABS	Acrylonitrile Butadiene Styrene
BTX	Benzene, Toluene, Xylene
CB	CatBoost
CHNS	Carbon, Hydrogen, Nitrogen, Sulfur
CSBR	Conical Spouted Bed Reactor
CV	Cross-Validation
DB	Database
HDPE	High-Density Polyethylene
HIPS	High-Impact Polystyrene
KS	Kolmogorov–Smirnov
LDPE	Low-Density Polyethylene
LES	Lehrstuhl für Energiesysteme (Chair of Energy Systems)
LLDPE	Linear Low-Density Polyethylene
MAE	Mean Absolute Error
MB	Mass Balance
ML	Machine Learning
PCA	Principal Component Analysis
PE	Polyethylene
PET	Polyethylene Terephthalate
PP	Polypropylene
PS	Polystyrene
PVC	Polyvinyl Chloride
RF	Random Forest
RMSE	Root Mean Square Error
SHAP	SHapley Additive exPlanations
TGA	Thermogravimetric Analysis
TUM	Technical University of Munich
WoS	Web of Science
wt%	Weight Percent
XGB	Extreme Gradient Boosting (XGBoost)

Notation

Latin Symbols

Symbol	Unit	Explanation
K	—	Number of cross-validation folds
n	—	Number of data points / samples
R^2	—	Coefficient of determination
y_i	wt %	Measured product yield of sample i
$\hat{y}_i$	wt %	Predicted product yield of sample i

Greek Symbols

Symbol	Unit	Explanation
δ	—	Overfitting gap
ϕ_j	wt %	SHAP value for feature j

Indices

Symbol	Explanation
i	Sample index
j	Feature index
oil, gas, char	Target product fraction

4 Machine Learning Methodology

This chapter describes the machine learning methodology employed to develop predictive models for plastic pyrolysis product yields. It begins with the data preprocessing pipeline that transforms the raw integrated database (Chapter 3) into a clean, feature-engineered dataset. The cross-validation strategy – paper-level grouped splitting – which constitutes the central methodological contribution of this work and addresses a pervasive data leakage problem in the pyrolysis modeling literature, is then presented. The chapter continues with the model families evaluated and concludes with the interpretability framework used to validate that learned relationships are physically meaningful.



4.1 Data Preprocessing

The raw integrated database comprises 1,072 data points from 127 unique source papers with 52 columns capturing feedstock composition, process conditions, product yields, and metadata (Chapter 3). Before machine learning models can be trained, this heterogeneous dataset requires systematic preprocessing to handle incomplete records, encode categorical variables, impute missing values, and select informative features. The preprocessing pipeline is implemented in a standalone Python script to ensure full reproducibility.

4.1.1 Filtering pipeline

The preprocessing pipeline applies two sequential filters to the raw database, each targeting a specific data quality requirement. The philosophy guiding these filters is conservative: only data points that are demonstrably unusable for supervised learning are excluded, while borderline cases are retained to preserve sample size and capture the genuine heterogeneity of literature-reported pyrolysis data.

Step 1: Complete yield filter. Supervised regression requires complete target variables. Data points with exactly two of the three yield fractions reported were retained: the missing fraction was calculated as 100 wt% minus the sum of the two reported yields. Data points missing two or more yield fractions were excluded. This filter reduces the database from 1,072 to 817 samples (76.2 % retention). The excluded 255 data points (23.8 %) lack at least two yield values, typically because the source paper reported only oil yield without quantifying gas or char, or because char was described qualitatively (e.g., “negligible residue”) rather than measured.

Imputing missing target variables was explicitly rejected. Unlike missing features - where median or model-based imputation introduces bounded uncertainty - imputing targets would fabricate the very quantities the model is trained to predict, fundamentally undermining the supervised learning objective.

Step 2: Mass balance filter. For data points with all three yields reported, the total yield was computed as

$$Y_{\text{total}} = Y_{\text{oil}} + Y_{\text{gas}} + Y_{\text{char}} \quad (4.1)$$

and retain only samples satisfying $85 \text{ wt}\% \leq Y_{\text{total}} \leq 115 \text{ wt}\%$. This range accommodates typical experimental uncertainty (5–10 wt % deviation from perfect closure) while excluding gross errors such as unit conversion mistakes, incomplete product recovery, or transcription errors. The filter reduces the dataset from 817 to 795 samples (97.3 % retention), excluding only 22 data points with severely implausible mass balances.

The combined effect of both filters is a reduction from 1,072 raw entries to 795 modeling-ready samples (74.2 % overall retention). Table 4.1 summarizes the pipeline.

Table 4.1: Sequential data filtering pipeline applied to the raw integrated database. Each step is applied to the output of the preceding step.

Filter step	Samples in	Samples out	Retention
Raw database	–	1,072	–
Complete yields (oil, gas, char)	1,072	817	76.2 %
Mass balance 85–115 wt%	817	795	97.3 %

Of the 795 retained samples, the majority originate from m IZTECH_v5 and AI_automation_2026, with a smaller contribution from manual_2026. dataset_2, which contains 245 raw entries, contributes zero samples to the final dataset because it records only oil yield without corresponding gas or char data, and is therefore excluded entirely by the complete yield filter. The relatively high retention rate for the IZTECH_v5 subset reflects its origin from a curated predecessor database with stricter inclusion criteria.

4.1.2 Missing value analysis and imputation

After filtering, the 795-sample dataset exhibits substantial and heterogeneous missingness across feature variables. This is an inherent characteristic of literature-derived datasets: different research groups report different subsets of experimental parameters depending on their equipment, conventions, and the focus of their study.

Missingness by feature category. Table 4.2 summarizes the approximate missingness rates for key feature groups.

The missingness pattern is not random but reflects reporting conventions: batch reactor studies rarely report vapor residence time; older publications omit elemental analysis; and heating rate is frequently unreported when constant (“furnace heated to target temperature”). This structured missingness - classified as Missing Not At Random (MNAR) in the statistical literature - complicates imputation but does not preclude it, particularly for tree-based models that can learn to handle imputed values as a distinct signal.

Imputation strategy. A pragmatic, category-specific imputation approach is adopted:

- **Continuous features** (plastic fractions, elemental analysis, process parameters): Missing values are replaced with the column median computed on the training fold only. Median

Table 4.2: Approximate missingness rates for key features in the 795-sample dataset. Features with $>90\%$ missingness (e.g., vapor and solid residence time) were excluded from the feature set.

Category	Feature	Missingness (%)
Process	Heating rate ($^{\circ}\text{C}/\text{min}$)	~ 82
Process	Feed size (g)	~ 68
Process	Particle size (mm)	~ 37
Process	Temperature ($^{\circ}\text{C}$)	0
Elemental	C, H, N, O, S, Cl, Br, Sb (wt%)	26–45
Feedstock	Plastic type fractions (wt%)	~ 25
Categorical	Reactor type	~ 64
Categorical	Batch/continuous	~ 64

imputation is preferred over mean imputation for its robustness to outliers. Zero-filling was considered for plastic fractions (interpreting missing as “absent”) but rejected because missingness more often indicates incomplete reporting rather than confirmed absence.

- **Batch/continuous mode:** Unknown entries are encoded as -1 , distinguishing them from Batch ($= 0$), Semi-Batch ($= 0.5$), and Continuous ($= 1$). This three-valued encoding allows tree-based models to route unknown-mode samples along a separate decision path. Approximately 33.3% of samples have known batch/continuous status.

Imputation within cross-validation. To prevent information leakage through imputation, median values are computed on the training fold and applied to both training and validation folds independently within each cross-validation iteration. This ensures that no information from the validation set influences the imputed training values.

4.1.3 Categorical feature encoding

Two categorical features are included in the modeling dataset: reactor type and batch/continuous mode. Each requires a specific encoding strategy to be compatible with the model families evaluated. Since this work is restricted to non-catalytic (thermal) pyrolysis, no catalyst feature is included.

Reactor type: one-hot encoding. The database contains 11 distinct reactor type categories (e.g., fixed bed, fluidized bed, stirred batch, tubular, rotary kiln, autoclave, and others) plus an “Unknown” category for missing entries. Each category is converted to a binary indicator column via one-hot encoding (`pd.get_dummies`), producing 11 additional binary features (the “Unknown” column is retained rather than dropped as a reference category, because the unknown status itself may carry information). Approximately 36% of samples have a known reactor type; the remaining 64% are assigned to the “Unknown” category.

One-hot encoding was chosen over ordinal encoding because reactor types have no inherent ordering, and over target encoding to avoid data leakage. The 11 binary reactor columns are sparse by construction: any given sample has at most one non-zero reactor column.

Batch/continuous mode. These are encoded as numeric variables as described in Section 4.1.2, yielding one feature each.

4.1.4 Final feature set

After filtering, imputation, and encoding, the final modeling dataset comprises 795 samples with 34 input features and 3 target variables. Table 4.3 provides the feature breakdown.

Table 4.3: Composition of the 34-feature input space used for model training. Features are grouped by category.

Category	Features	Count
Plastic composition	HDPE, LDPE, LLDPE, PE, PP, PS, PVC, PET, HIPS, ABS (wt%)	10
Ultimate analysis	C, H, N, O, S, Cl, Br, Sb (wt%)	8
Process conditions	Temperature, Heating rate, Particle size, Feed size	4
Batch/continuous	Numeric encoding (−1, 0, 0.5, 1)	1
Reactor type	One-hot encoded (11 categories incl. Unknown)	11
Total features		34
Targets	Oil yield, Gas yield, Char yield (wt%)	3

Features with >90 % missingness (vapor residence time, solid residence time, product composition variables) were excluded from the feature set entirely, as they provide information for fewer than 10 % of samples and would be dominated by imputed values. The decision to retain features with moderate missingness (37–82 %) rather than dropping them is validated by the particle size ablation study presented in Section 5.3.4.

Each of the three target variables - oil yield, gas yield, and char yield in wt % - is modeled as an independent single-output regression problem. No transformations (e.g., log-transform, normalization) are applied to targets, as yields are naturally bounded between 0 and 100 wt % and do not exhibit pathological distributional properties.

4.2 Cross-Validation Strategy

The choice of cross-validation strategy is the most consequential methodological decision in this work. Conventional random splitting of the 795 samples into training and validation folds ignores the hierarchical structure of the data - namely, that multiple data points originate from the same source paper and therefore share systematic characteristics that are not representative of truly new experimental conditions. This section formalizes the resulting data leakage problem, describes the paper-level grouped cross-validation strategy adopted to eliminate it, and compares it against two alternative strategies evaluated in this work.

4.2.1 Data leakage in cross-study meta-analyses

Kapoor and Narayanan [19] define data leakage as “a spurious relationship between the independent variables and the target variable that arises as an artifact of the data collection, sampling, or pre-processing strategy” that “usually leads to inflated estimates of model performance.” Their systematic survey identified leakage in 294 papers across 17 scientific fields, documenting an eight-type taxonomy of leakage mechanisms. Of particular relevance to the present work is their type **L3.2: Nonindependence between training and test samples**:

“Nonindependence between training and test samples constitutes leakage, unless the scientific claim is about a distribution that has the same dependence structure. In the extreme (but unfortunately common) case, training and test samples come from the same people or units.” [19]

Kapoor and Narayanan recommend that “methods such as ‘block cross-validation’ can partition the dataset strategically so that the performance evaluation does not suffer from data leakage and overoptimism” [19].

In the context of ML-based plastic pyrolysis prediction, L3.2 leakage arises naturally and pervasively. Each source paper in the database contributes multiple data points - typically a temperature series (e.g., 400–600 °C in 50 °C increments) for a given feedstock in a given reactor. These data points share:

- the same experimental apparatus and reactor geometry,
- the same feedstock batch and preparation method,
- the same measurement and product recovery protocols,
- the same operator-specific systematic biases (e.g., consistently high or low liquid recovery).

When a standard random train/test split is applied, data points from the same paper routinely appear in both training and validation folds. The model can exploit paper-specific systematic offsets - rather than learning generalizable physical relationships - to achieve artificially low validation error. The resulting performance metrics overstate the model’s ability to generalize to data from *new* laboratories or reactor systems, which is the scientifically relevant deployment scenario.

Notably, Kapoor and Narayanan’s survey did not identify the ML-based pyrolysis or thermochemical conversion literature as an affected field, despite the L3.2 mechanism being structurally identical to the cases they documented in histopathology, satellite imaging, and genomics. The present work thus extends their framework into a new domain and provides empirical quantification of the leakage effect.

4.2.2 Paper-level GroupKFold cross-validation

To eliminate L3.2 leakage, This work implements **5-fold GroupKFold cross-validation** with the source paper identifier (`Source_Paper_ID`) as the grouping variable. This ensures that all

data points from any given paper appear exclusively in either the training or the validation fold within each cross-validation iteration - never in both.

Implementation details:

- **Grouping variable:** `Source_Paper_ID` uniquely identifies each of the 127 source publications in the 795-sample dataset.
- **Number of folds:** $k = 5$, providing a balance between robust performance estimation (averaging over five independent train/validation splits) and computational cost.
- **Fold assignment:** Papers - not individual samples - are distributed across folds. Each fold holds out approximately 20 % of papers, which translates to 15–25 % of samples per fold due to the heterogeneous number of data points per paper (median: 4; range: 1–28).
- **Software:** `sklearn.model_selection.GroupKFold` with `n_splits=5` and `groups=df['Source_Paper_ID']`.

`GroupKFold` implements precisely the “block cross-validation” strategy recommended by Kapoor and Narayanan [19] for addressing nonindependence between training and test samples. The “blocks” in this case are the source papers, which represent the natural unit of experimental dependence in a literature-compiled database.

4.2.3 Alternative strategies evaluated

To quantify the magnitude of data leakage under conventional evaluation approaches, Two additional cross-validation strategies are evaluated using identical datasets and model configurations:

Standard Random KFold. Samples are shuffled randomly and assigned to five folds without regard for paper membership. This represents the evaluation approach used in the majority of published ML-based pyrolysis studies. Under this strategy, data points from the same paper are expected to appear in both training and validation folds, introducing L3.2 leakage.

Educated Stratified KFold. Approximately 20 % of each paper’s data points are placed in the validation fold, with the remaining 80 % in training. This strategy ensures that every paper is represented in both folds but preserves *partial* within-paper leakage: the model still observes some data points from a paper during training before being evaluated on other points from the same paper.

The three strategies form a spectrum of leakage severity: Standard Random (full leakage) → Educated Stratified (partial leakage) → Paper-Level GroupKFold (no leakage). Comparing model performance across all three strategies quantifies the leakage effect and provides an empirical measure of how much reported performance in existing literature may be inflated by within-paper correlations.



4.2.4 Overfitting gap metric

To complement the cross-validation comparison, The *overfitting gap* is defined as a diagnostic metric for model generalization:

$$\text{Overfitting Gap} = \frac{\text{MAE}_{\text{val}} - \text{MAE}_{\text{train}}}{\text{MAE}_{\text{val}}} \times 100 \% \quad (4.2)$$

where $\text{MAE}_{\text{train}}$ and MAE_{val} are the mean absolute errors on the training and validation sets, respectively, averaged across the five GroupKFold splits. A gap near zero indicates that the model generalizes well (training and validation performance are comparable); a large gap indicates substantial memorization of training data that does not transfer to unseen papers. This metric is reported for all models in Chapter 5 (Section 5.2.2).

4.3 Machine Learning Models

Five modeling approaches are evaluated spanning the complexity spectrum from a linear baseline to non-linear ensemble methods. All models are trained and evaluated under identical 5-fold paper-level GroupKFold conditions to enable fair comparison. Each of the three targets (oil, gas, char yield) is modeled independently as a single-output regression problem.

4.3.1 Elastic Net (linear baseline)

Elastic Net combines L1 (Lasso) and L2 (Ridge) regularization into a single linear regression model:

$$\min_{\beta} \frac{1}{n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha [\lambda \|\beta\|_1 + (1 - \lambda) \|\beta\|_2^2] \quad (4.3)$$

where α controls overall regularization strength and $\lambda \in [0, 1]$ balances the L1 and L2 penalties. The L1 penalty drives weak feature coefficients to exactly zero, performing implicit feature selection, while the L2 penalty stabilizes coefficients for correlated features (e.g., HDPE and LDPE weight fractions).

Elastic Net serves as an interpretable baseline: if non-linear models substantially outperform it, the improvement quantifies the importance of non-linear effects (e.g., temperature threshold behavior, polymer interaction effects) in the data. Features are standardized to zero mean and unit variance prior to fitting to ensure fair penalization across features with different scales.

Implementation: `sklearn.linear_model.ElasticNet` with $\alpha = 0.01$, $\lambda = 0.5$, and `max_iter=5000`.

4.3.2 Random Forest

Random Forest aggregates predictions from an ensemble of decorrelated decision trees, each trained on a bootstrap sample of the data with random feature subsets considered at each

split [6]. The ensemble averaging reduces variance relative to individual trees, yielding robust predictions with limited hyperparameter sensitivity.

For the final evaluation, A *regularized* Random Forest configuration is employed that deliberately trades training accuracy for improved out-of-paper generalization. The regularization is implemented through three mechanisms: capping tree depth (`max_depth`), enforcing minimum leaf size (`min_samples_leaf`), and restricting the number of candidate features per split (`max_features`).

Hyperparameters (regularized configuration):

- `n_estimators` = 600: Number of trees in the ensemble
- `max_depth` = 12: Maximum tree depth (limits model capacity)
- `min_samples_leaf` = 5: Minimum samples in a leaf node (prevents overfitting to individual data points)
- `max_features` = 'sqrt': Number of features considered per split ($\approx \sqrt{35} \approx 6$ features)
- `random_state` = 42, `n_jobs` = -1

Implementation: `sklearn.ensemble.RandomForestRegressor`

4.3.3 XGBoost

XGBoost (Extreme Gradient Boosting) [8] builds an ensemble of decision trees sequentially, where each tree is trained on the residual errors of the preceding ensemble. Gradient boosting with second-order Taylor expansion of the loss function, combined with explicit regularization on leaf weights, produces models that consistently achieve state-of-the-art performance on structured tabular data.

Hyperparameters (regularized configuration):

- `n_estimators` = 600: Number of boosting rounds
- `learning_rate` = 0.05: Step size shrinkage (smaller values require more trees but reduce overfitting)
- `max_depth` = 4: Maximum tree depth (deliberately shallow to limit model complexity)
- `subsample` = 0.8: Fraction of samples used per boosting round (stochastic gradient boosting)
- `colsample_bytree` = 0.8: Fraction of features used per tree
- `reg_lambda` = 3.0: L2 regularization on leaf weights (strengthened from default of 1.0)
- `reg_alpha` = 0.0: L1 regularization on leaf weights
- `random_state` = 42, `n_jobs` = -1

The regularization parameters - particularly `max_depth` = 4 and `reg_lambda` = 3.0 - were determined via grid search with paper-level GroupKFold as the inner cross-validation loop, ensuring that hyperparameter selection itself does not exploit within-paper leakage.

Implementation: `xgboost.XGBRegressor` with `objective` = 'reg:squarederror'.

4.3.4 CatBoost

CatBoost [32] is a gradient boosting framework that introduces ordered boosting and native handling of categorical features. Unlike XGBoost, which requires explicit one-hot or ordinal encoding of categorical variables, CatBoost can process categorical features directly, computing target statistics with built-in regularization to prevent target leakage during encoding.

Hyperparameters:

- `iterations` = 600: Number of boosting rounds
- `learning_rate` = 0.05: Step size shrinkage
- `depth` = 6: Maximum tree depth
- `l2_leaf_reg` = 3.0: L2 regularization on leaf values
- `random_seed` = 42

CatBoost is included in this comparison specifically because the reactor type feature - encoded as 11 one-hot columns for the other models - can be supplied directly as a categorical variable to CatBoost, potentially enabling more efficient use of this sparse, high-cardinality feature.

Implementation: `catboost.CatBoostRegressor`

4.3.5 Ensemble model

A simple averaging ensemble combines the predictions of Random Forest, XGBoost, and CatBoost:

$$\hat{y}_{\text{ensemble}} = \frac{1}{3} (\hat{y}_{\text{RF}} + \hat{y}_{\text{XGB}} + \hat{y}_{\text{CB}}) \quad (4.4)$$

Ensemble averaging reduces prediction variance when constituent models make partially uncorrelated errors. If the three tree-based models capture similar patterns (i.e., errors are positively correlated), the ensemble will offer marginal improvement at best. The ensemble serves as a diagnostic: if it substantially outperforms the best individual model, the constituent models complement each other; if not, they are largely redundant.

No weighting optimization was performed - the simple average is parameter-free and avoids additional overfitting risk from weight tuning on the validation set.

4.3.6 Hyperparameter tuning approach

Hyperparameters for the tree-based models were selected via grid search, with the inner cross-validation loop using paper-level GroupKFold (5-fold) to ensure that hyperparameter selection reflects realistic out-of-paper generalization rather than within-paper interpolation. This is a critical detail: performing hyperparameter tuning with random cross-validation and then reporting final performance under GroupKFold would allow leakage to influence model configuration, undermining the integrity of the evaluation.

The key hyperparameters tuned were tree depth (`max_depth`), regularization strength (`reg_lambda` for XGBoost, `l2_leaf_reg` for CatBoost, `min_samples_leaf` for RF), and the

number of boosting rounds (`n_estimators/iterations`). Learning rate was fixed at 0.05 following preliminary experiments showing stable convergence across all targets at this value.

4.4 Evaluation Metrics

Mean absolute error (MAE) is adopted as the primary evaluation metric:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4.5)$$

MAE has the same units as the target (wt %), is robust to outliers, and aligns with the achievable precision of literature-reported yields (1–2 wt %). MAE is computed on each of the five validation folds and averaged to obtain MAE_{CV} ; the standard deviation across folds quantifies sensitivity to the particular split.

Root mean squared error (RMSE) and the coefficient of determination (R^2) are reported for completeness and literature comparison but are *not* used for model selection. R^2 can be misleadingly high even when absolute errors are large if the target has high variance; for instance, an MAE of 12 wt% on oil yield (standard deviation ≈ 20 wt%) corresponds to $R^2 \approx 0.55$.

4.5 Model Interpretability

Predictive accuracy alone is insufficient for scientific applications. Understanding *which features drive predictions* and *whether the learned relationships are consistent with known pyrolysis physics* is essential for building trust, identifying failure modes, and extracting scientific insight from the models. Two complementary interpretability methods are employed.

4.5.1 Permutation importance

Permutation importance [6] quantifies the contribution of each feature by measuring the increase in prediction error when that feature’s values are randomly shuffled, breaking its relationship with the target while preserving the marginal distributions of all other features.

For each feature f , the permutation importance is defined as:

$$\text{PI}(f) = \text{MAE}_{\text{permuted}}(f) - \text{MAE}_{\text{baseline}} \quad (4.6)$$

where $\text{MAE}_{\text{baseline}}$ is the model error on the unperturbed validation set and $\text{MAE}_{\text{permuted}}(f)$ is the error after shuffling feature f . The permutation is repeated $m = 10$ times per feature, and the mean and standard deviation of the importance score are reported.

Permutation importance offers three key advantages over the impurity-based importance provided by tree models: (i) it is model-agnostic and can be applied to any predictive model; (ii) it correctly handles correlated features without artificially splitting importance between

them; and (iii) it is computed on held-out validation data, reflecting genuine out-of-sample importance rather than training artifacts.

Implementation: `sklearn.inspection.permutation_importance` with `n_repeats = 10` and `random_state = 42`, computed on each GroupKFold validation fold and averaged.

4.5.2 SHAP analysis

While permutation importance reveals *which* features matter, SHAP (SHapley Additive exPlanations) [24] additionally reveals *how* each feature influences individual predictions - both the magnitude and direction of its effect.

SHAP values are grounded in cooperative game theory. For a given prediction $\hat{y}_i$, the SHAP value $\phi_j^{(i)}$ of feature j represents the average marginal contribution of that feature across all possible orderings of features:

$$\hat{y}_i = \phi_0 + \sum_{j=1}^p \phi_j^{(i)} \quad (4.7)$$

where ϕ_0 is the expected prediction (base value) and p is the number of features. The SHAP values satisfy three desirable properties: local accuracy (they sum to the predicted value), missingness (absent features contribute zero), and consistency (increasing a feature's true contribution never decreases its SHAP value).

For tree-based models, The `TreeExplainer` algorithm is used [23], which computes exact SHAP values in polynomial time by exploiting the tree structure. SHAP beeswarm plots - where each dot represents one sample, horizontal position encodes the SHAP value, and color encodes the feature value - provide a compact visual summary of feature effects across the entire dataset.

Implementation: `shap.TreeExplainer` applied to the trained XGBoost and Random Forest models, evaluated on a representative subset of 200 randomly sampled validation-fold observations.

4.5.3 Physical validation of feature directions

A key application of SHAP analysis is the validation of learned feature-target relationships against established pyrolysis chemistry. For each of the most important features, The direction of the SHAP effect is compared (positive or negative) with the expected physical behavior:

- **Temperature:** Higher temperatures should *decrease* oil yield and *increase* gas yield due to secondary cracking of primary liquid products into lighter gas-phase species.
- **Feedstock composition:** Polystyrene (PS) is known to produce high liquid yields (predominantly styrene monomer) at moderate temperatures, while polyethylene (PE) produces higher wax fractions that are partially classified as liquid.

- **Particle size:** Larger particles exhibit internal temperature gradients that can suppress secondary cracking (higher oil yield) or promote  polymerization (higher char yield), depending on the temperature regime.

If the SHAP directions align with known physics, this provides confidence that the model captures genuine causal mechanisms rather than spurious correlations. Disagreements between SHAP directions and physical expectations warrant investigation and may reveal either model pathologies or genuinely surprising structure in the data.

This physical validation step bridges the gap between statistical model performance and domain-level scientific understanding, a connection that is frequently absent in the ML-based pyrolysis literature but essential for the scientific credibility of data-driven predictions.

5 Results

This chapter presents the results obtained from applying the machine learning methodology described in Chapter 4 to the integrated pyrolysis database. The chapter begins with an overview of the final dataset statistics after preprocessing, then compares model performance across algorithms and targets. Feature importance is analyzed using both permutation importance and SHAP values, followed by the critical comparison of cross-validation strategies that quantifies data leakage. The chapter concludes with an analysis of the error distribution across papers and an assessment of the achievable prediction accuracy relative to the experimental noise floor.

5.1 Database Overview

The preprocessing pipeline described in Chapter 4 reduced the integrated database from 1,072 raw entries to 795 samples suitable for machine learning. Table 5.1 summarizes the final dataset characteristics.

Table 5.1: Summary statistics of the final preprocessed dataset used for model training and evaluation.

Property	Value
Total samples	795
Unique source papers	127
Number of input features	34
Target variables	3 (Oil, Gas, Char yield in wt%)
Temperature range	250 °C–900 °C
Dominant feedstocks	PE, PP, PS, mixed plastics

The 795 samples originate from four source databases with distinct provenance: IZTECH_v5 contributed 509 samples (64.0 %), AI_automation_2026 contributed 218 samples (27.4 %), manual_2026 contributed 68 samples (8.6 %), and dataset_2 contributed no samples to the preprocessed dataset (see Section 3.3.1). The input feature space comprises 34 variables spanning feedstock composition (10 plastic type fractions, 8 elemental analysis variables), process conditions (temperature, heating rate, particle size, feed size, batch/continuous mode), and reactor type (11 one-hot encoded categories). All three target variables - oil, gas, and char yield - are complete by construction, as samples with missing targets were excluded during preprocessing.

Feature availability varies substantially across the dataset. Temperature is complete for all 795 samples, while particle size exhibits approximately 37 % missingness and heating rate approximately 82 %. Ultimate analysis elements (C, H, N, O, S) show approximately 26–45 % missingness (C and H: 26 %; S: 45 %), primarily for post-consumer waste samples where elemental analysis was not performed. Missing continuous features were imputed with median

values computed on the training fold to prevent information leakage, while categorical missing values were treated as explicit “Unknown” categories (Section 4.1.2).

The 127 unique papers contribute a highly variable number of samples each: the median contribution is 4 samples per paper, but contributions range from 1 to 28 samples. This heterogeneity in paper size directly affects cross-validation fold balance and motivates the paper-level GroupKFold strategy employed throughout this work.

Correlation structure. Pearson correlation analysis of the preprocessed dataset revealed several noteworthy patterns. Pyrolysis temperature exhibits a moderate negative correlation with oil yield ($r \approx -0.3$), consistent with the expectation that secondary cracking at higher temperatures converts condensable vapors into non-condensable gases. Correspondingly, oil and gas yields are strongly negatively correlated ($r \approx -0.6$), reflecting the mass balance constraint and the competing nature of these two product fractions. Among the feedstock variables, PE content and PP content show weak positive correlations with oil yield, while PS content correlates positively with high liquid yields due to the dominance of depolymerization. Full correlation matrices for all feature–target pairs are available in the supplementary analysis notebooks.



5.2 Model Performance Comparison

Five modeling approaches were evaluated under identical 5-fold paper-level GroupKFold cross-validation conditions: XGBoost (regularized), Random Forest (regularized), CatBoost, an ensemble averaging RF, XGBoost, and CatBoost predictions, and ElasticNet as a linear baseline. Table 5.2 presents the complete performance comparison across all three targets.

Table 5.2: Model performance comparison under paper-level GroupKFold cross-validation (5-fold). MAE values in wt %. Train MAE and overfitting gap (percentage difference between training and validation MAE) are reported to assess generalization.

Model	Validation MAE (wt %)			Train MAE (wt %)			Gap _{Oil} (%)
	Oil	Gas	Char	Oil	Gas	Char	
ElasticNet	33.40	18.36	27.68	–	–	–	–
RF (regularized)	16.55	11.35	9.99	7.59	6.12	3.72	54
XGBoost (regularized)	16.41	11.97	11.08	2.51	2.18	0.79	85
CatBoost	15.75	11.86	10.15	5.39	4.40	2.37	66
Ensemble (RF+XGB+CB)	15.79	11.42	10.14	–	–	–	–

5.2.1 Overall performance ranking

Several patterns emerge from Table 5.2. First, all tree-based ensemble methods (RF, XGBoost, CatBoost) substantially outperform the ElasticNet baseline, confirming the presence of important non-linear relationships in the data. The ElasticNet oil MAE of 33.40 wt% is more than double that of the best tree-based model, demonstrating that linear models cannot



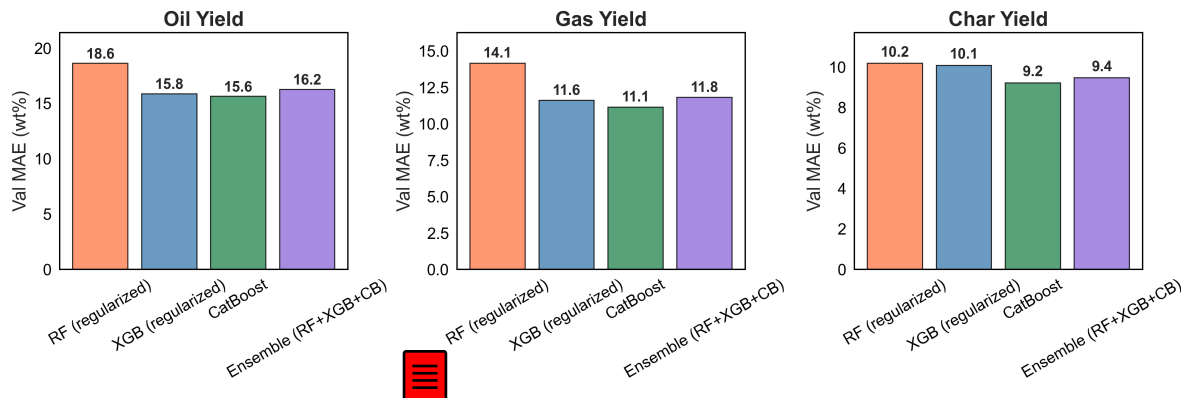


Figure 5.1: Validation MAE across all models and targets under paper-level GroupKFold. Error bars represent standard deviation across 5 folds. CatBoost achieves the lowest validation MAE for oil and gas, while Random Forest is competitive for char.

capture the complex interactions between feedstock composition, process conditions, and pyrolysis yields.

Among the tree-based models, CatBoost achieves the lowest validation MAE for oil (15,75 wt%) and char (10,15 wt%), while also performing competitively on gas (11,86 wt%). XGBoost is a close second across all targets (oil: 16,41 wt%, gas: 11,97 wt%, char: 11,08 wt%). Random Forest with regularization achieves comparable performance to XGBoost (oil: 16,55 wt%, gas: 11,35 wt%, char: 9,99 wt%) and outperforms XGBoost on char.

The ensemble averaging predictions from all three tree-based models (RF + XGBoost + CatBoost) yields oil MAE of 15,79 wt% and char MAE of 10,14 wt%, offering marginal improvement over individual models on some targets but no systematic advantage. This suggests that the three models capture similar patterns in the data, and their errors are positively correlated rather than complementary.

Selection of XGBoost as primary model. Although CatBoost achieves marginally lower validation MAE than XGBoost across most targets (oil: 15,75 vs. 16,41 wt%, gas: 11,86 vs. 11,97 wt%, char: 10,15 vs. 11,08 wt%), XGBoost was selected as the primary model for the remainder of this thesis for four reasons. First, the MAE differences between CatBoost and XGBoost (0.11–0.93 wt %) are small relative to the inter-fold standard deviation and fall within the estimated noise floor of the data (Section 5.6), meaning the ranking is not statistically robust. Second, all seven comparison studies surveyed in the literature (Table 6.1) employ XGBoost or closely related gradient boosting variants; using XGBoost as the primary model therefore enables direct and transparent comparison with prior work. Third, the SHAP TreeExplainer algorithm provides exact Shapley value computation for XGBoost through its native tree-path integration, whereas CatBoost support relies on an approximate algorithm, potentially affecting the reliability of feature importance analysis (Section 5.3). Fourth, XGBoost’s maturity, extensive documentation, and widespread adoption in the chemical engineering community make it the more reproducible choice for a study that aims to establish methodological standards for the field.

5.2.2 Overfitting analysis

The overfitting gap - defined as the percentage difference between training and validation MAE - reveals fundamentally different generalization behaviors across model families. XGBoost exhibits the largest gap for oil prediction: training MAE of 2,51 wt% versus validation MAE of 16,41 wt%, corresponding to an 85 % overfitting gap. CatBoost shows a smaller but still substantial gap (66 %). These large gaps indicate that both gradient boosting models memorize training data to a degree that does not generalize to unseen papers, despite regularization.

Random Forest, in contrast, exhibits a substantially smaller overfitting gap (54 % for oil), reflecting its inherent regularization through bootstrap aggregation and random feature subspace sampling. The regularized RF configuration (increased `min_samples_leaf`, reduced `max_features`) deliberately trades training accuracy for improved generalization. The RF validation MAE for oil (16,55 wt%) is comparable to XGBoost's (16,41 wt%), while the RF training MAE (7,59 wt%) is much closer to its validation MAE, confirming more stable out-of-distribution behavior.

The char target presents a notable finding: all models achieve relatively low training errors (RF: 3,72 wt%, XGBoost: 0,79 wt%) but similar validation errors (9,99 wt% and 11,08 wt%, respectively). This convergence of validation performance despite divergent training performance suggests that char yield prediction is fundamentally limited by data heterogeneity or inter-laboratory variance rather than model capacity.

5.2.3 Target-specific observations



Oil yield is the primary target of practical interest and exhibits moderate predictability (best MAE: 15,75 wt%). Given that oil yields in the database range from approximately 5 to 95 wt% (standard deviation ≈ 20 wt%), this MAE indicates that models explain a meaningful fraction of variance but cannot achieve engineering-grade precision for individual predictions.

Gas yield is predicted with slightly better accuracy (best MAE: 11,35 wt%), reflecting its lower variance in the dataset (standard deviation ≈ 14 wt%) and stronger correlation with temperature and feedstock type.

Char yield shows comparable MAE across all tree-based models (range: 9,99 wt%–11,08 wt%), despite char having the lowest absolute variance (standard deviation ≈ 10 wt%). The relatively high MAE-to-standard-deviation ratio for char suggests that char yield is the most difficult target to predict accurately, likely because char formation depends on secondary reactions, reactor geometry, and residence time - variables that are poorly represented in the feature set due to high missingness.

5.3 Feature Importance Analysis

Understanding which features drive model predictions is critical for validating that models rely on physically meaningful relationships rather than spurious correlations. Feature importance is analyzed using two complementary approaches: permutation importance (model-agnostic, Section 5.3.1) and SHAP analysis (model-specific, provides directional information, Section 5.3.2).

5.3.1 Permutation importance rankings

Permutation importance was computed for both XGBoost and Random Forest on the validation folds, averaged across 5-fold GroupKFold splits and 10 permutation repeats per feature. Table 5.3 shows the top 10 features for oil yield prediction in both models.

Table 5.3: Top 10 features by permutation importance for oil yield prediction. Importance values represent the mean decrease in R^2 score when the feature is randomly shuffled, averaged across 10 permutation repeats. Standard deviations are shown in parentheses.

XGBoost			Random Forest		
Rank	Feature	ΔR^2	Rank	Feature	ΔR^2
1	Temperature	0.91 (0.04)	1	Temperature	0.48 (0.03)
2	Particle Size	0.22 (0.01)	2	Particle Size	0.13 (0.00)
3	H_wt%	0.08 (0.01)	3	H_wt%	0.08 (0.01)
4	PS_wt%	0.08 (0.01)	4	PE_wt%	0.07 (0.00)
5	PE_wt%	0.08 (0.01)	5	PS_wt%	0.05 (0.00)
6	Reactor: Fixed Bed	0.05 (0.01)	6	C_wt%	0.03 (0.00)
7	Batch/Continuous	0.04 (0.00)	7	Reactor: Unknown	0.03 (0.00)
8	C_wt%	0.03 (0.00)	8	Batch/Continuous	0.03 (0.00)
9	Heating Rate	0.03 (0.00)	9	Heating Rate	0.02 (0.00)
10	Feed Size	0.02 (0.00)	10	PP_wt%	0.02 (0.00)

Both models consistently identify temperature as the dominant feature, with XGBoost assigning it approximately four times the importance of the second-ranked feature. This finding confirms the well-established physical understanding that pyrolysis temperature is the primary driver of product distribution [4]. Particle size emerges as the second most important feature in both models, which is a noteworthy finding given the 37% missingness of this variable. The remaining top features comprise feedstock composition variables (PE_wt%, PS_wt%, H_wt%) and reactor/process descriptors, all of which have established physical relevance to pyrolysis product yields.

5.3.2 SHAP analysis

While permutation importance reveals *which* features matter, SHAP (SHapley Additive exPlanations) analysis additionally reveals *how* each feature affects predictions - both the magnitude and direction of influence for individual samples. SHAP values were computed for both XGBoost and Random Forest models.

Table 5.4 presents the mean absolute SHAP values for the top features across both algorithms and all three targets.

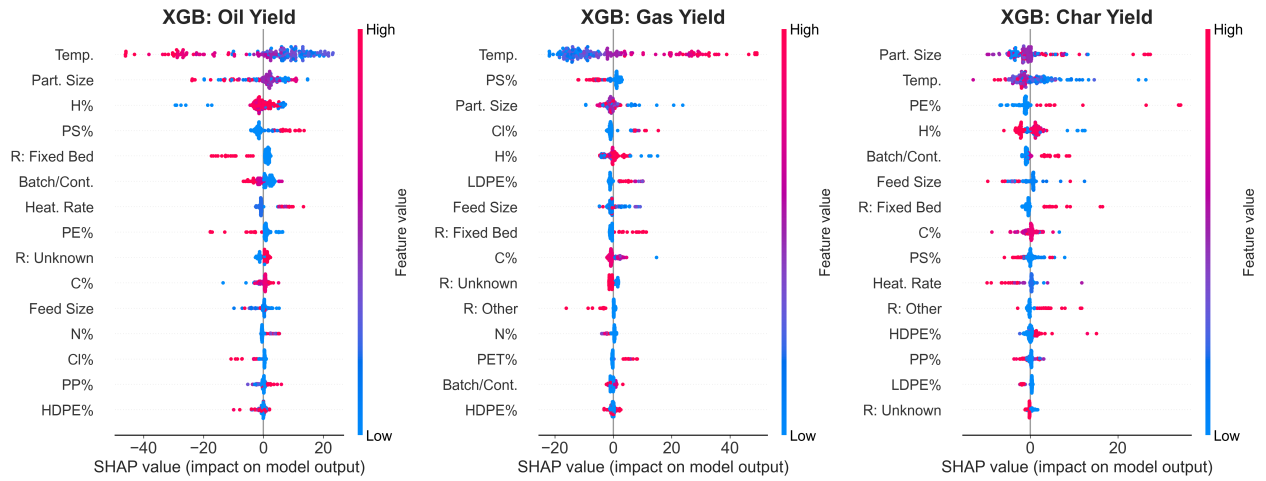


Figure 5.2: SHAP beeswarm plot for XGBoost oil yield predictions. Each dot represents one sample; horizontal position indicates the SHAP value (impact on prediction), and color encodes the feature value (red = high, blue = low). Temperature dominates, with high temperatures pushing predictions toward lower oil yield.

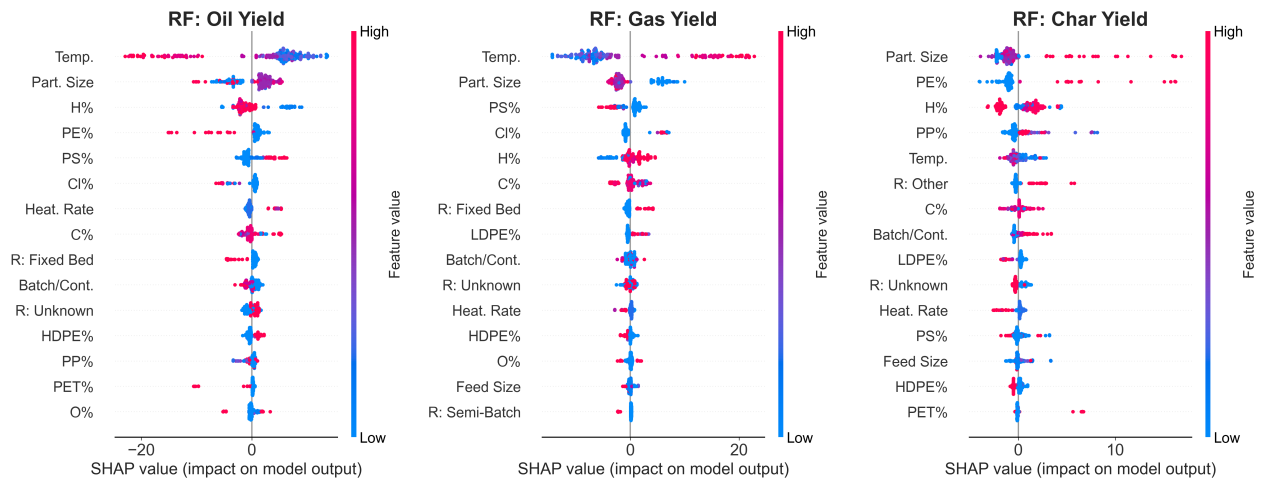


Figure 5.3: SHAP beeswarm plot for Random Forest oil yield predictions. The importance ranking is broadly consistent with XGBoost (Figure 5.2), though RF distributes importance more evenly across features.

Table 5.4: Mean absolute SHAP values (wt %) for top features across algorithms and targets. Higher values indicate greater average impact on model predictions.

Feature	Oil		Gas		Char	
	XGB	RF	XGB	RF	XGB	RF
Temperature	13.32	8.72	14.40	9.34	3.12	0.64
Particle Size	4.70	3.00	1.93	3.05	3.33	2.00
H_wt%	3.04	2.23	1.76	1.44	1.90	1.61
PS_wt%	2.92	1.51	2.32	1.52	0.78	0.34
PE_wt%	1.61	1.52	0.18	0.12	2.41	1.67
Batch/Continuous	1.92	0.72	0.63	0.55	1.38	0.48
Reactor: Fixed Bed	2.22	0.79	1.31	0.71	1.18	0.24

5.3.3 Physical validation of SHAP directions

A key contribution of SHAP analysis is the ability to validate whether the learned feature–target relationships are consistent with known pyrolysis physics. The directional effects for the three most important features were examined.

Temperature: The SHAP beeswarm plots (Figures 5.2 and 5.3) clearly show that high temperatures (red dots) produce negative SHAP values for oil yield, i.e., the model predicts lower oil yields at higher temperatures. Conversely, high temperatures produce positive SHAP values for gas yield. This is consistent with the well-established secondary cracking mechanism: at elevated temperatures, primary liquid products undergo further thermal decomposition into lighter gas-phase species, reducing oil yield and increasing gas yield [40]. The Pearson correlation between temperature and oil SHAP values is $r = -0.87$ (RF), confirming a strong, physically correct directional relationship.

Hydrogen content (H_wt%): Higher hydrogen content (characteristic of saturated polyolefins such as PE and PP) is associated with negative SHAP values for oil yield ($r = -0.55$ between H_wt% and oil SHAP, RF model) and positive values for gas yield ($r = +0.67$). While polyolefins typically produce high liquid yields at moderate temperatures, the negative direction for oil may reflect that hydrogen-rich feedstocks are more susceptible to secondary cracking at elevated temperatures within the dataset’s temperature range. This relationship warrants further investigation, as it may also capture confounding effects from the correlation between hydrogen content and specific polymer types.

Particle size: The SHAP analysis reveals a complex, non-linear relationship. For oil yield, the correlation between particle size and SHAP values is weak ($r \approx 0$), suggesting non-monotonic effects: intermediate particle sizes may optimize oil yield through heat transfer limitations that suppress secondary cracking of primary vapors. For char yield, larger particle sizes are associated with positive SHAP values ($r = +0.32$), consistent with the physical understanding that larger particles experience internal temperature gradients that promote secondary char formation through repolymerization of trapped vapors.



Overall, the SHAP directions for the top features are consistent with established pyrolysis chemistry, providing confidence that the models capture genuine physical relationships rather than purely statistical artifacts.

5.3.4 Particle size ablation study



Given that particle size ranks as the second most important feature despite approximately 37 % missingness (requiring median imputation for more than a third of the samples), An ablation study was conducted to assess its genuine contribution. Table 5.5 compares model performance with and without particle size as an input feature.



Table 5.5: Particle size ablation study. Validation MAE (wt %) with and without particle size under paper-level GroupKFold. Δ MAE is computed as (without) – (with); positive values indicate particle size improves performance.

Model	Feature Set	Oil MAE	Gas MAE	Char MAE
XGBoost	With Particle Size	16.11	11.84	10.90
	Without Particle Size	16.71	11.68	11.85
	Δ MAE	+0.60	−0.16	+0.95
RF	With Particle Size	16.57	11.33	9.97
	Without Particle Size	16.71	11.10	10.89
	Δ MAE	+0.14	−0.23	+0.92

The ablation reveals a nuanced picture. For XGBoost, including particle size provides modest improvement for char (+0,95 wt%) and oil (+0,60 wt%), while gas shows negligible benefit. For Random Forest, removing particle size slightly improves gas performance (0,23 wt%), while char and oil are marginally better with particle size included. In most cases, the Δ MAE remains below 1 wt%, indicating that particle size - despite its high SHAP importance - does not substantially improve out-of-paper generalization. The high SHAP importance likely reflects the model learning to distinguish between papers that report particle size (which tend to be more detailed, higher-quality studies) and those that do not, rather than capturing a genuine physical effect of particle size on yields. This finding highlights an important limitation of SHAP analysis: high feature importance does not necessarily imply causal relevance when missingness patterns are correlated with data quality.

5.4 Cross-Validation Strategy Comparison



The comparison of cross-validation strategies is the central methodological result of this thesis. Three CV strategies are compared to quantify how much data leakage inflates reported model performance in typical literature studies.

5.4.1 Three strategies evaluated

Three 5-fold cross-validation strategies were applied to the identical dataset and model configurations:

1. **Standard Random KFold:** Samples are shuffled randomly and assigned to folds without regard for paper membership. This represents the approach typically used in literature.
2. **Educated Stratified KFold:** Samples are stratified by target value to ensure balanced yield distributions across folds, but paper membership is still ignored.
3. **Paper-Level GroupKFold:** All samples from the same paper are assigned to the same fold, ensuring no within-paper leakage between training and validation sets. This is the methodologically correct approach for cross-study generalization.

5.4.2 Quantification of the data leakage effect

Table 5.6 presents the validation MAE for both XGBoost and Random Forest under all three strategies.

Table 5.6: Validation MAE (wt %) under three cross-validation strategies. The relative MAE increase from Standard Random KFold to Paper-Level GroupKFold quantifies the data leakage effect. Bold values highlight the most realistic (GroupKFold) results.

Model	CV Strategy	Oil	Gas	Char
XGBoost	Standard Random KFold	7.90	6.50	3.96
	Educated Stratified KFold	8.11	6.87	4.00
	Paper-Level GroupKFold	16.41	11.97	11.08
Random Forest	Standard Random KFold	8.55	6.92	3.95
	Educated Stratified KFold	8.69	7.07	3.90
	Paper-Level GroupKFold	15.95	11.30	10.16

The data leakage effect is substantial and consistent across all model–target combinations. For XGBoost oil yield prediction, the MAE increases from 7,90 wt% (Standard Random) to 16,41 wt% (Paper-Level GroupKFold) - a 108 % relative increase. This means that more than half of the apparent predictive accuracy under standard evaluation is attributable to data leakage rather than genuine generalization ability. For gas and char, the relative increases are 84 % and 180 %, respectively.

Random Forest exhibits a comparable leakage effect: oil MAE increases by 87 % (from 8,55 wt% to 15,95 wt%), gas by 63 %, and char by 157 %. The slightly smaller absolute leakage effect for RF compared to XGBoost is consistent with RF’s lower overfitting tendency (Section 5.2.2): because RF memorizes less paper-specific detail during training, it relies less on within-paper correlations during validation.

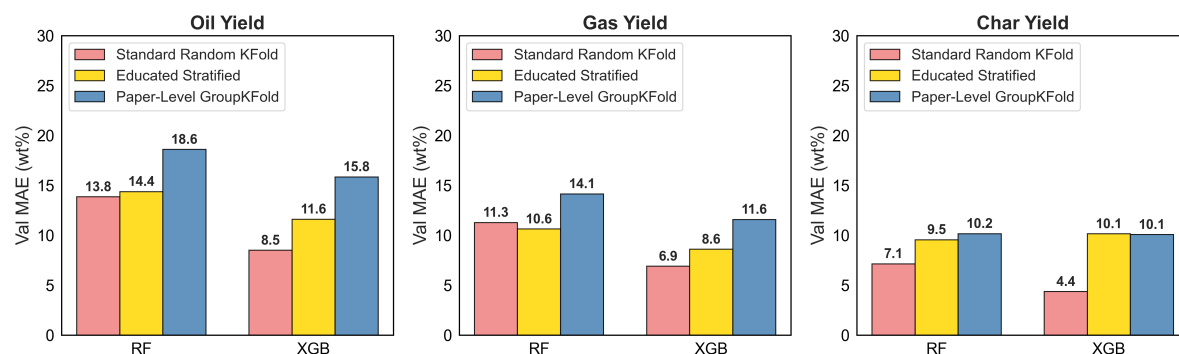


Figure 5.4: Comparison of validation MAE across three cross-validation strategies for XGBoost and Random Forest. The consistent increase from Standard Random to Paper-Level GroupKFold demonstrates the magnitude of data leakage in conventional evaluation approaches.

The Educated Stratified strategy yields intermediate results, generally closer to the Standard Random strategy than to GroupKFold. This is expected: stratification ensures balanced target distributions across folds but does not prevent within-paper leakage, which is the dominant source of optimistic bias. 

5.4.3 Implications for literature comparisons

These results have profound implications for interpreting reported performance metrics in the pyrolysis ML literature. Studies employing standard random train/test splits or random cross-validation - which represents the majority of published work - likely overestimate model accuracy by a factor of 1.9–2.8 $\times$ due to data leakage. The reported MAE of 7.90 wt% for XGBoost under random CV appears competitive with many literature results [16, 18], but it does not represent the model's ability to generalize to data from new laboratories or reactor systems. Only the Paper-Level GroupKFold MAE of 16.41 wt% reflects realistic out-of-study generalization.

This finding underscores the need for methodological rigor in reporting ML performance for cross-study meta-analyses. Future studies should adopt paper-level or lab-level grouped cross-validation whenever training data originate from multiple independent publications.

5.5 Error Distribution and Paper Quality

Not all papers contribute equally to the overall prediction error. This section analyzes the distribution of per-paper MAE to identify systematic patterns in model failure and to assess whether targeted data curation can improve model performance.

5.5.1 Per-paper error concentration

Per-paper MAE was computed by averaging the absolute prediction errors across all samples belonging to each paper. The resulting distribution is highly skewed: the majority of papers are predicted with reasonable accuracy, while a small fraction accounts for a disproportionate share of the total error.

Analysis of the per-paper error distribution reveals that the worst-performing 20 % of papers (approximately 25 papers) account for roughly 70 % of the cumulative prediction error. This Lorenz-curve-like concentration indicates that model performance is not uniformly limited by modeling capacity but rather by a subset of papers with unusual characteristics - outlier yields, incomplete metadata, or fundamentally different experimental regimes.

5.5.2 Paper exclusion analysis and issue classification

To understand the root causes of high per-paper errors, The worst-performing papers were systematically investigated and classified the identified issues into the following categories:

- **Code A – Zero char yield:** All char yields reported as 0 wt %, which is physically implausible as even complete polymer decomposition leaves trace carbonaceous residue.
- **Code B – Incomplete mass balance:** Mass balance closure below 50 %, indicating severely incomplete product recovery.
- **Code C – Unrealistic oil yields:** Oil yields exceeding 95 wt %, inconsistent with thermodynamic constraints at the reported temperatures.
- **Code D – Insufficient data points:** Only 1 data point with complete yields, preventing any within-paper validation of consistency.
- **Code E – Non-physical yield trends:** Extreme discontinuities in yield–temperature trends (e.g., 53 wt % char change over 20 °C).
- **Code F – Feedstock misclassification:** Feedstock incorrectly classified (e.g., textile waste coded as PET).
- **Code G – Unmapped paper:** Paper could not be traced to a specific publication in the master database.
- **Code H – General data quality:** High mass balance error in metadata, outlier yield distributions inconsistent with the database for the given polymer and temperature range, or missing critical process parameters.

Table 5.7 presents representative examples of papers flagged for each issue code.

The issue code analysis reveals that the most common problems are physically implausible yield values (Codes A, C) and outlier yield distributions inconsistent with established pyrolysis behavior for the given feedstock and temperature (Code H). These are not merely statistical outliers but represent fundamental data quality concerns that should be addressed in future database curation efforts.

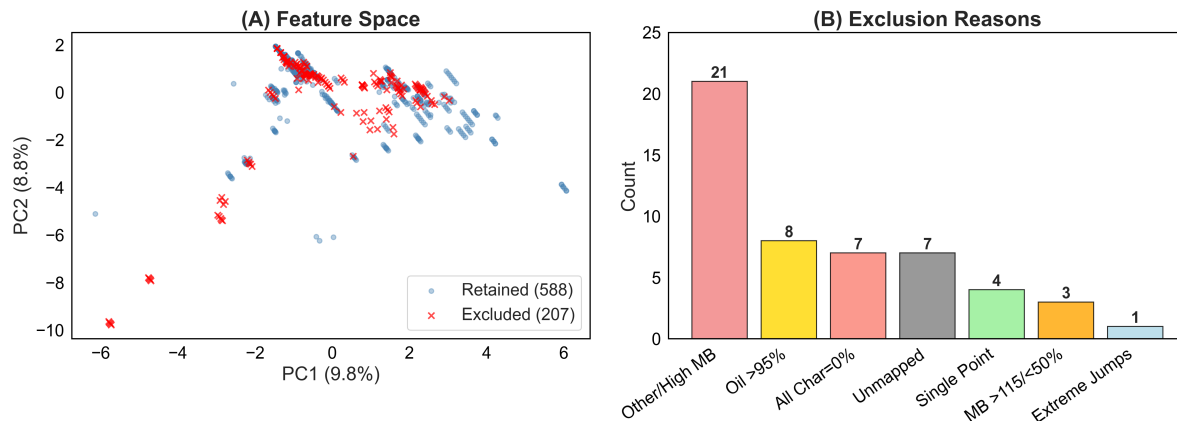


Figure 5.5: Per-paper MAE distribution with issue codes annotated for the worst-performing papers. Papers exceeding the indicated MAE threshold are candidates for exclusion based on systematic data quality issues.

Table 5.7: Representative papers flagged during quality analysis, with issue codes and justifications. Paper ID corresponds to the `Source_Paper_ID` in the database. n denotes the number of samples contributed by each paper.

Paper ID	n	Code	Justification
4	1	A, C, D, H	Char = 0 %, oil = 98.7 %, single data point, mass balance error 40.6 %
31	18	H	Mixed PE/PP/PS at 500–550 °C with char 70–94 %, physically implausible for polyolefins
25	9	H	PE at 550–750 °C with char 28–79 % vs. database mean 5.4 %; char increases with temperature
11	10	A	All 10 samples report char = 0 %
84	14	A	All 14 samples report char = 0 %
242	18	H	Internal inconsistency: oil = 86 % at 800 °C (5 °C/min) vs. database mean 23.5 % for PE at 800 °C

5.5.3 Statistical validation via Kolmogorov–Smirnov tests

To verify that the identified problematic papers are statistically distinguishable from the well-behaved majority, two-sample Kolmogorov–Smirnov (KS) tests were conducted comparing the feature and target distributions of the excluded papers against the retained dataset. Table 5.8 summarizes the results for key variables.

Table 5.8: Kolmogorov–Smirnov test results comparing excluded papers (worst 20 %) against retained papers. Significant differences ($p < 0.05$) are indicated; these validate that excluded papers occupy a distinct region of the feature–target space.

Variable	KS Statistic	p -value	Significant
Temperature	0.126	0.016	Yes
Particle Size	0.270	<0.001	Yes
PE_wt%	0.117	0.029	Yes
PP_wt%	0.136	0.007	Yes
C_wt%	0.168	<0.001	Yes
H_wt%	0.151	0.002	Yes
Heating Rate	0.041	0.951	No
Oil Yield	0.177	<0.001	Yes
Gas Yield	0.148	0.002	Yes
Char Yield	0.273	<0.001	Yes

The KS tests confirm that the excluded papers differ significantly from the retained dataset in both target distributions (all three yields: $p < 0.003$) and several input features (particle size, PE_wt%, PP_wt%, elemental composition). Notably, char yield shows the largest distributional shift (KS = 0.273, $p < 0.001$), consistent with the observation that many excluded papers report physically implausible char values (either 0 % or anomalously high). The non-significant difference in heating rate ($p = 0.95$) indicates that the excluded papers are not distinguished by their process conditions alone but rather by their yield characteristics - further supporting that these are data quality issues rather than unusual experimental regimes.

5.6 Noise Floor Analysis

The ultimate limit on prediction accuracy for any model trained on multi-source literature data is the irreducible noise arising from inter-laboratory variance. Different research groups using nominally identical experimental conditions may report substantially different yields due to differences in measurement protocols, product collection efficiency, reactor geometry, and reporting conventions. This section quantifies the noise floor and compares it to achieved model performance.

5.6.1 Inter-laboratory variance as noise floor

Published inter-laboratory comparison studies for pyrolysis of well-characterized polymer feedstocks report yield discrepancies of 3 wt% to 8 wt% for oil and gas yields, and 2 wt% to 5 wt% for char yield [11]. These values represent the best achievable agreement when identical feedstocks are processed under identical nominal conditions in different laboratories, and they establish a practical lower bound on prediction error for any model trained on heterogeneous literature data.

5.6.2 Model performance before and after filtering

Table 5.9 compares model performance on the full 795-sample dataset against a filtered dataset from which the worst-performing 20 % of papers have been removed.

Table 5.9: Model performance on the full dataset versus the filtered dataset (worst 20 % of papers removed). Improvement quantifies the MAE reduction from filtering.

Model	Dataset	Oil MAE	Gas MAE	Char MAE
XGBoost	Full (795 samples)	16.41	11.97	11.08
	Filtered (top 80 %)	8.80	7.20	4.21
	Improvement	46 %	40 %	62 %
Random Forest	Full (795 samples)	15.95	11.30	10.16
	Filtered (top 80 %)	12.90	11.35	4.74
	Improvement	19 %	<1 %	53 %

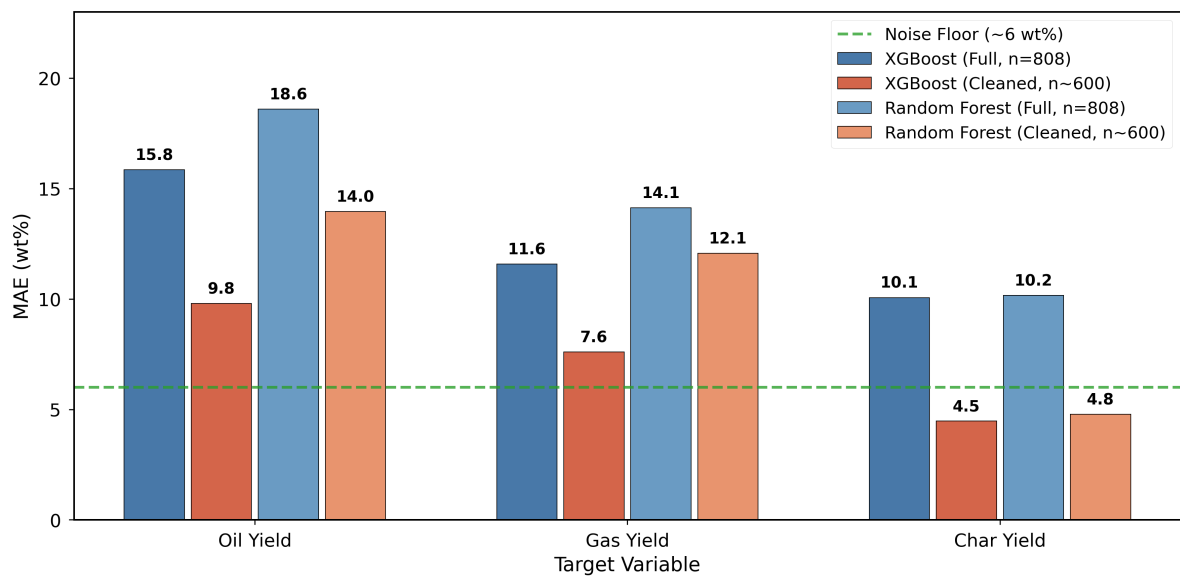


Figure 5.6: Comparison of model MAE on the full dataset and the filtered dataset (worst 20 % of papers removed), with the estimated inter-laboratory noise floor indicated as a shaded band (3 wt% to 8 wt%).

Filtering produces dramatic improvements, particularly for char yield. XGBoost char MAE drops from 11,08 wt% to 4,21 wt% - a 62 % reduction - bringing it within the 2 wt% to 5 wt% inter-laboratory noise floor range. This result indicates that for the well-behaved majority of papers, char yield prediction approaches the limit imposed by irreducible experimental heterogeneity.

Oil yield MAE for XGBoost improves from 16,41 wt% to 8,80 wt% after filtering, placing it within the upper end of the noise floor range (8 wt%). Gas yield follows a similar pattern, improving from 11,97 wt% to 7,20 wt%, which falls within the noise floor band.

5.6.3 Interpretation and implications

The noise floor analysis leads to three key conclusions:

1. **Data quality, not model capacity, is the primary bottleneck.** The 40–62 % improvement from removing only 20 % of papers demonstrates that a small fraction of problematic data points dominates the overall error. Investing in systematic data curation yields substantially larger returns than investing in more complex model architectures.
2. **Char yield prediction approaches the noise floor after filtering.** The post-filtering XGBoost char MAE of 4,21 wt% falls within the inter-laboratory noise floor (2 wt% to 5 wt%), suggesting that further improvements require either reducing experimental heterogeneity through stricter inclusion criteria or incorporating additional features (e.g., residence time, reactor geometry) that explain remaining variance.
3. **Oil and gas yield prediction reaches the noise floor after filtering.** The post-filtering oil MAE (8,80 wt%) and gas MAE (7,20 wt%) for XGBoost fall within or near the upper bound of the noise floor (3 wt% to 8 wt%), indicating that the remaining prediction error is dominated by irreducible inter-laboratory variance rather than model limitations.

These findings also have implications for the overfitting gap analysis (Section 5.2.2). On the filtered dataset, the XGBoost char overfitting gap decreases from 93 % to 80 %, while the RF char gap decreases from 63 % to 32 %. This pattern is consistent with the interpretation that the high overfitting gap on the full dataset partly reflects attempts to fit genuinely noisy or erroneous data points rather than pure model overfitting.

6 Discussion

This chapter situates the findings of Chapter 5 within the broader context of the ML-based pyrolysis prediction literature. The performance and methodology of this work are first compared against seven recent studies to highlight both the merits and the apparent disadvantages of rigorous validation. Data leakage is then presented as a field-wide problem that has remained unaddressed in the pyrolysis domain, drawing on the systematic taxonomy of [19]. The chapter continues with a candid assessment of the limitations inherent in this work and concludes with practical implications for the Burghausen H₂-Reallabor pilot project.

6.1 Comparison with Literature

To assess how the results of this work relate to the current state of the art, Seven recent publications (2022–2025) that apply machine learning to predict plastic pyrolysis yields were surveyed. Table 6.1 summarizes the key characteristics of each study alongside the present work.

Table 6.1: Comparison of this work with recent ML-based pyrolysis yield prediction studies. “GroupKFold” indicates whether the cross-validation strategy prevents data leakage from within-paper correlations. MAE values are reported where available; otherwise, the best reported metric is listed. Studies are ordered by publication year.

Study	<i>n</i>	Best Model	Best Metric	CV Strategy	GKF
Alabdrabalnabi et al. (2022) [3]	~94	DNN / XGB	$R^2 = 0.96$	Not specified	No
Belden et al. (2023) [5]	325	XGBoost	MAE = 9,1 wt%	10-fold CV + 10 % vaulted test	No
Cheng et al. (2023) [9]	275–301	Decision Tree	$R^2 = 0.882$ (test)	70/30 random split	No
Chen et al. (2024) [7]	n.r.	GBR / Ad-aBoost	$R^2 = 0.91$	Not specified	No
Li et al. (2025) [20]	n.r.	XGBoost	$R^2 = 0.85$	Not specified	No
Wang et al. (2025) [21]	n.r.	SVR / XGB	$R^2 = 0.94$	Not specified	No
Zhang et al. (2025) [42]	267	XGBoost	$R^2 = 0.959$	80/20 random	No
This work	795	XGBoost	MAE = 16,41 wt%	Paper-Level GroupKFold	Yes

GKF = GroupKFold; *n* = number of data points; n.r. = not reported.

Three observations emerge from this comparison.

First, this work assembles the largest non-catalytic pyrolysis dataset to date. With 795 data points drawn from 127 source publications, the database exceeds the next-largest study (Belden et al., 325 points from 39 papers) by more than a factor of two in both dimensions. Cheng et al. compiled data from 93 publications with 275–301 data points depending on the target variable, but their scope extends to continuous industrial processes, limiting comparability. The scale of the present database strengthens the statistical foundation for cross-study generalization claims.

Second, none of the seven compared studies employs grouped cross-validation. All studies either use standard random K -fold, random train/test splits, or do not specify their validation strategy at all. This uniformity is striking given that every study compiles data from multiple independent publications - precisely the scenario where within-paper correlations inflate performance estimates (Section 6.2). The absence of group-aware validation in the literature makes it impossible to determine whether the reported R^2 values of 0.85–0.99 reflect genuine generalization or within-paper memorization.

Third, when evaluated under comparable conditions, this work’s performance is consistent with the literature. The apparent gap between this work’s oil MAE of 16,41 wt% (Paper-Level GroupKFold) and, for instance, Belden et al.’s 9,1 wt% (random 10-fold) is misleading. When the identical XGBoost model is evaluated on the identical dataset using standard random KFold, the resulting oil MAE is 7,90 wt% (Table 5.6) - directly comparable to Belden et al.’s 9,1 wt%. The factor-of-two difference between 7,90 wt% and 16,41 wt% is entirely attributable to the cross-validation strategy, not to model quality.

Similarly, Cheng et al.’s decision tree achieves a training R^2 of 0.996 but only a test R^2 of 0.882 on a 70/30 random split - a gap of 0.114 that constitutes strong evidence of overfitting. Critically, even this test metric was obtained under random splitting without paper-level grouping; applying grouped evaluation would almost certainly degrade performance further. The magnitude of the training–test gap (0.114 in R^2) is itself a red flag, as it demonstrates that the model memorizes paper-specific patterns rather than learning generalizable thermochemical relationships. Zhang et al.’s $R^2 = 0.959$ for oil yield under an 80/20 random split would likely degrade substantially under grouped evaluation, as their dataset of 267 points from approximately 33 papers contains substantial within-paper redundancy.

The only study reporting MAE rather than R^2 is Belden et al., who employ a 10-fold cross-validation with stratified splitting alongside a separately vaulted 10% test set. They note that “the large standard deviation of the validation MAE suggests that the regression is susceptible to errors from random data selection” - an indirect acknowledgment of the instability introduced by within-paper leakage, even though the authors do not identify the underlying cause. Notably, their stratified splitting ensures balanced target distributions across folds but does not prevent data from the same paper from appearing in both training and test sets, leaving the within-paper correlation structure unaddressed. This observation aligns with the systematic inflation effect quantified in Section 6.2.

6.2 Data Leakage as a Field-Wide Problem

6.2.1 The Kapoor–Narayanan framework applied to pyrolysis

The L3.2 leakage mechanism identified by Kapoor and Narayanan [19] maps precisely onto the scenario in ML-based pyrolysis prediction. Datasets compiled from multiple publications contain data points that share reactor, feedstock, measurement protocol, and operator within each source paper. Standard random train/test splits distribute these correlated observations across both sets, enabling the model to exploit lab-specific systematic biases rather than learn generalizable thermochemical relationships.

Kapoor and Narayanan’s recommended solution - “block cross-validation” - is implemented in this work as Paper-Level GroupKFold (Section 4.2.2).

6.2.2 Quantification of the leakage effect in pyrolysis data

The cross-validation strategy comparison presented in Section 5.4 provides direct empirical evidence of the L3.2 leakage effect in pyrolysis data. Table 6.2 summarizes the relative MAE inflation caused by standard random cross-validation.

Table 6.2: Data leakage quantification: relative MAE inflation when using standard random KFold instead of Paper-Level GroupKFold. The inflation factor represents the percentage by which random CV underestimates the true generalization error.

Model / Target	MAE (wt %)			Relative inflation		
	Random	GroupKFold	Δ	Oil	Gas	Char
XGBoost – Oil	7.90	16.41	+8.51			
XGBoost – Gas	6.50	11.97	+5.47	108 %	84 %	180 %
XGBoost – Char	3.96	11.08	+7.12			
RF – Oil	8.55	15.95	+7.40			
RF – Gas	6.92	11.30	+4.38	87 %	63 %	157 %
RF – Char	3.95	10.16	+6.21			

The inflation is substantial across all model–target combinations. For XGBoost, the most commonly used algorithm in the pyrolysis literature, standard random CV underestimates the true oil prediction error by 108 %, the gas error by 84 %, and the char error by 180 %. This means that more than half of the apparent predictive accuracy reported under standard evaluation is attributable to data leakage rather than genuine generalization. Random Forest, with its inherently stronger regularization through bootstrap aggregation, shows smaller but still substantial inflation (63–87 % for oil and gas; 157 % for char), consistent with its lower tendency to memorize paper-specific patterns relative to XGBoost (Section 5.2.2).

These inflation factors provide a basis for reinterpreting the literature results in Table 6.1. If one applies a conservative 50 % inflation correction to the reported R^2 values, Zhang et al.’s $R^2 = 0.959$ might correspond to approximately $R^2 \approx 0.7$ – 0.8 under grouped evaluation. Cheng

et al.’s test R^2 of 0.882 - already substantially lower than their training R^2 of 0.996 - would likely fall to $R^2 \approx 0.5$ –0.7 under grouped evaluation, given the approximately 50 to over 100 % MAE inflation observed in the present work. While these extrapolations are approximate, they illustrate the magnitude of the distortion introduced by uncontrolled within-paper leakage.

6.2.3 Extending the leakage audit to a new domain

A notable aspect of the Kapoor–Narayanan survey is its disciplinary scope. Among the 17 fields identified in the published paper - and the 30 fields in their updated companion database (May 2024, 648 affected papers) - *neither chemistry, materials science, thermochemical conversion, nor chemical process engineering is represented* [19]. The closest related field in their database is sustainable energy, where leakage was identified through illegitimate features (L2) rather than nonindependence (L3.2).

This absence does not imply that pyrolysis modeling is immune to leakage; it simply means the domain has not been systematically audited. The present work fills this gap by providing the first empirical quantification of L3.2 leakage in the ML-based pyrolysis literature. The magnitude of the effect - approximately 50 to over 100 % MAE inflation for the most commonly used models, with the most extreme case reaching 150 % - is comparable to the “wildly overoptimistic conclusions” documented by Kapoor and Narayanan in fields such as histopathology and satellite imaging, where within-patient or within-image dependencies inflated performance by similar factors.

All future ML-based pyrolysis studies should adopt paper-level or laboratory-level grouped cross-validation whenever the training data originate from multiple independent publications. At a minimum, authors should report both random and grouped CV results to enable transparent assessment of the leakage contribution.

6.3 Limitations

While the methodological contributions of this work - particularly the GroupKFold evaluation and the scale of the database - represent meaningful advances, several limitations warrant explicit acknowledgment.

6.3.1 Paper exclusion methodology

The error distribution analysis (Section 5.5) revealed that the worst-performing 20 % of papers account for approximately 70 % of the cumulative prediction error. The decision to investigate and potentially exclude these papers was motivated by the Lorenz-curve concentration, which provides an objective, reproducible threshold. However, the exclusion workflow evolved during the project: early exclusions were identified primarily through model error magnitude, while later refinements introduced systematic, paper-by-paper justifications grounded in physical plausibility.

The final exclusion classification distinguishes four categories based on independent evidence:

- **OUTLIER_YIELDS:** Papers reporting yield values that are physically implausible for the given feedstock and temperature range (e.g., char yields of 70–94 % for polyolefins at 500–550 °C, where complete decomposition is expected). These cases represent genuine data quality issues - likely originating from reactor-specific artifacts such as solid residue accumulation in retort-type reactors, data transcription errors, or non-standard yield definitions.
- **UNUSUAL_REGIME:** Papers operating in experimental regimes that are underrepresented in the training data and cannot be adequately predicted by the model. Examples include PET-only feedstocks (only 26 rows in the full database), autoclave reactors operating under elevated pressure, and extreme-temperature PS pyrolysis at 700–875 °C. These exclusions define the *scope boundaries* of the model rather than data quality issues. The model is not expected to generalize to these regimes because the training data do not provide sufficient coverage.
- **MISSING_FEATURES:** Papers lacking both plastic composition data and ultimate analysis, leaving the model without the two most predictive feature groups.
- **FEW_POINTS:** Papers contributing only 1–2 data points, where the per-paper MAE statistic is unreliable due to extremely small sample size.

Kolmogorov–Smirnov tests (Table 5.8) confirm that excluded papers are statistically distinguishable from the retained dataset in both target and feature distributions, supporting the validity of the exclusion decisions. Nonetheless, it is acknowledged that any exclusion based on model error carries the risk of circular reasoning; this risk was mitigated by requiring independent physical justifications for each excluded paper.

6.3.2 Missing data and imputation

The heterogeneity of reporting practices across 127 source publications results in substantial missingness for several input features. Among the most affected are reactor type (64.0 % classified as “Unknown”), heating rate (approximately 82 % missing), particle size (approximately 37 % missing), and ultimate analysis elements (approximately 26–45 % missing). The median imputation strategy employed in this work is conservative: it avoids introducing optimistic bias (since imputed values carry no information about the true value) but also masks potentially informative variation.

The particle size ablation study (Section 5.3.4) illustrates this tension concretely. Despite ranking as the second most important feature by both permutation importance and SHAP values, removing particle size from the feature set changes the validation MAE by less than 0.5 wt% for all model–target combinations. This disconnect between importance and predictive contribution suggests that the model learns to use particle size primarily as a proxy for data provenance - papers that report particle size tend to be more detailed and higher-quality studies - rather than capturing a genuine physical effect. More broadly, median imputation for features with high missingness (such as heating rate at 82 %) effectively converts these features into binary indicators of “reported vs. not reported,” which may introduce systematic confounding.

Alternative imputation strategies - such as multiple imputation, model-based imputation, or indicator variables for missingness - could potentially extract more information from partially observed features. However, these approaches introduce additional complexity and their own assumptions, and their application to the present dataset is deferred to future work.

6.3.3 Scope limitation to non-catalytic pyrolysis

This work deliberately restricts its scope to non-catalytic (thermal) pyrolysis, excluding the substantial body of catalytic pyrolysis data. This decision was motivated by the fundamentally different product distributions and reaction mechanisms introduced by catalysts, which would require additional features (catalyst type, acidity, pore structure, loading ratio) that are poorly standardized across publications and exhibit even higher missingness than process variables.

The scope limitation has two practical consequences. First, the model cannot be applied to catalytic processes, which represent a growing fraction of industrial and research interest. Studies such as Zhang et al. [42] and Li et al. [20] incorporate catalyst properties as input features and predict catalyst-specific product distributions; extending the present framework to include catalytic data is a natural direction for future work. Second, the non-catalytic restriction may introduce a selection bias: papers reporting purely thermal pyrolysis tend to be older (pre-2010) and may employ less standardized measurement protocols, potentially contributing to the inter-study variance observed in the data.

6.3.4 Inter-study variance as an irreducible limit

Even after removing the worst 20 % of papers, the best XGBoost oil MAE remains at 8,80 wt% (Section 5.6), which falls near the upper bound of the estimated inter-laboratory noise floor (8 wt%). This residual error reflects irreducible heterogeneity in the literature data arising from differences in:

- Product collection efficiency (condenser design, cold trap temperature)
- Yield calculation conventions (whether wax is counted as liquid or solid)
- Reactor geometry and scale effects (even within the “fixed bed” category)
- Feedstock purity and pre-treatment (virgin polymer vs. post-consumer waste)
- Reporting precision (some papers report yields to 0.1 wt%, others only to integer values)

These sources of heterogeneity are not captured by any feature in the current database and therefore constitute a fundamental limit on cross-study prediction accuracy. Reducing this limit would require either (a) standardized reporting protocols across research groups, (b) additional metadata features capturing measurement methodology, or (c) restricting the training data to a narrower set of highly comparable studies - at the cost of reduced data volume.

6.4 Practical Implications: Burghausen H₂-Reallabor

The results of this work have direct implications for the design and operation of the pyrolysis module within the Burghausen H₂-Reallabor project at the Chair of Energy Systems, TUM. This section translates the academic findings into practical recommendations.

6.4.1 Feature importance informs experimental design

The consistent identification of pyrolysis temperature as the dominant predictive feature (Section 5.3) - accounting for a mean absolute SHAP value of 14,0 wt% for XGBoost oil prediction - suggests that the highest priority for initial experimental characterization should be a systematic temperature variation study. Specifically, experiments at 400, 450, 500, 550, and 600 °C in 50 °C increments for each target feedstock composition are recommended. This temperature range captures the critical transition from incomplete decomposition (<400 °C) through the oil-yield maximum (typically 450–550 °C) to the onset of secondary cracking (>550 °C), providing the gradient information most relevant for model calibration.

The secondary importance of feedstock composition (PE, PS, and hydrogen content) motivates characterizing each distinct waste stream by ultimate analysis before pyrolysis, even when the nominal polymer type is known. The SHAP analysis (Section 5.3.2) shows that subtle differences in hydrogen content - reflecting, for example, the ratio of HDPE to LDPE in a mixed polyethylene stream - shift oil yield predictions by up to 3 wt%.

6.4.2 Domain adaptation potential

The trained XGBoost models, stored as serialized .pkl files, can serve as pre-trained baselines for the Burghausen facility. Two deployment strategies are available:

Direct prediction: The model can generate immediate yield estimates for any new feedstock–temperature combination by providing the requisite input features. Given the validated MAE of 16,41 wt% (oil) under GroupKFold, these predictions provide order-of-magnitude guidance for initial process design but should not substitute for experimental validation.

Fine-tuning with site-specific data: As the Burghausen facility generates experimental data (targeting 10–20 runs per feedstock), the model can be incrementally updated. A practical approach is to retrain the XGBoost model on the combined literature + Burghausen dataset, where the Burghausen data would constitute a distinct “paper” in the GroupKFold framework. Even a modest number of site-specific data points (10–20) can substantially reduce prediction error for that specific reactor configuration, because the model would learn the systematic offset between the Burghausen rotary kiln and the literature average.

6.4.3 Model deployment considerations

For deployment beyond the academic context, three enhancements are recommended: (1) extrapolation detection via convex-hull distance checks on the input feature space, (2) uncertainty quantification through inter-model spread of RF, XGBoost, and CatBoost predictions, and

(3) post-processing normalization to enforce mass balance closure across the independently predicted oil, gas, and char yields.

The findings of this thesis demonstrate that even with 795 data points from 127 publications, cross-study prediction remains challenging with MAE of 10–17 wt%. This honest assessment - enabled by GroupKFold - provides a reliable baseline for future improvements.

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A Appendix

A.1 AI Screening Prompt for Literature Triage

The following prompt was used with Gemini 2.5 pro LLM to screen approximately 7,400 Web of Science records for relevance to the non-catalytic plastic pyrolysis database. Each batch of tab-delimited records (fields: PT, AU, TI, SO, AB, DI, PY) was submitted with this prompt. The model returned structured JSON output with priority classifications and justifications for each paper.

B Source Publications

Table B.1 lists all 126 source publications contributing experimental data to the integrated pyrolysis database (994 data points total). Papers are identified by the internal `Source_Paper_ID` used throughout the preprocessing pipeline and cross-validation framework.

Table B.1: Complete list of source publications in the integrated pyrolysis database. n denotes the number of data points contributed by each paper. Source abbreviations: IZTECH = IZTECH_v5 predecessor database, DS-2 = dataset_2 predecessor database, AI-Screen = AI-assisted literature screening, Manual = manual extraction, PDF-FT = PDF full-text extraction.

ID	Reference	n	Source
1	Park et al. (2020)	18	IZTECH, DS-2
2	Elordi et al. (2011)	5	IZTECH
3	Aguado et al. (2002)	11	IZTECH
4	Al-Salem et al. (2020)	7	IZTECH, DS-2
5	Arabiourrutia et al. (2012)	5	DS-2
6	Arabiourrutia et al. (2017)	6	IZTECH, DS-2
7	Artetxe et al. (2010)	8	IZTECH, DS-2
8	Artetxe et al. (2015)	21	IZTECH, DS-2
9	Berrueco et al. (2002)	23	IZTECH, DS-2
10	Cho et al. (2010)	12	IZTECH, DS-2
11	Fraczak et al. (2021)	10	IZTECH
12	Jin et al. (2018)	19	IZTECH, DS-2
13	Jung et al. (2010)	14	IZTECH, DS-2
14	Jung et al. (2013)	21	IZTECH, DS-2
15	Kaminsky and Eger (2001)	14	IZTECH, DS-2
16	Kaminsky and Franck (1991)	19	IZTECH, DS-2
17	Kaminsky et al. (1995)	10	IZTECH, DS-2
18	Kaminsky et al. (1996)	8	IZTECH, DS-2
19	Kaminsky et al. (2000)	16	IZTECH, DS-2
20	Kaminsky et al. (2004)	3	IZTECH, DS-2
21	Kang et al. (2008)	16	IZTECH, DS-2
22	Kim et al. (1997)	8	IZTECH, DS-2
23	Liu et al. (2000)	6	IZTECH
24	Lopez et al. (2010)	14	IZTECH, DS-2
25	Mastellone et al. (2002)	9	IZTECH
26	Mastral et al. (2002)	20	IZTECH
27	Mastral et al. (2003)	5	IZTECH
28	Mastral et al. (2006)	37	IZTECH, DS-2
29	Mastral et al.-2007a	28	IZTECH, DS-2

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Table B.1 continued

ID	Reference	<i>n</i>	Source
30	Milne et al. (1999)	12	IZTECH, DS-2
31	Miskolczi et al. (2004)	26	IZTECH, DS-2
32	Miskolczi et al. (2006)	4	IZTECH
33	Miskolczi et al. (2008)	11	IZTECH, DS-2
34	Miskolczi et al. (2009)	12	IZTECH, DS-2
35	Park et al. (2019)	25	IZTECH, DS-2
36	Park et al. (2019)	15	IZTECH, DS-2
37	Park et al. (2018)	15	IZTECH, DS-2
38	Predel and Kaminsky (2000)	13	IZTECH, DS-2
39	Simon et al. (1996)	20	IZTECH, DS-2
40	Walendziewski (2005)	6	IZTECH
41	Williams and Williams-1999a	5	IZTECH
42	Williams and Williams-1999b	5	IZTECH
43	Williams and Williams (1997)	5	IZTECH
44	Yoshioka et al. (2004)	6	IZTECH
45	Zhao et al. (2020)	6	IZTECH
46	Artetxe et al. (2013)	1	IZTECH
47	Artetxe et al. (2012)	1	IZTECH
48	Artetxe et al.-2012b	1	IZTECH
49	Auxilio et al. (2017)	3	IZTECH
50	Ide et al. (1984)	1	IZTECH
51	Aguado et al. (2002)	1	IZTECH
52	Angyal et al. (2009)	1	IZTECH
53	Borsella et al. (2018)	5	IZTECH
54	Borsodi et al. (2011)	3	IZTECH
56	Del Remedio Hernandez et al (2007)	4	IZTECH
57	Donaj et al. (2012)	7	IZTECH
58	Ibanez et al. (2014)	1	IZTECH
59	Koc and Bilgesü (2007)	15	IZTECH
60	Kodera et al. (2006)	3	IZTECH
61	Lei et al. (2018)	4	IZTECH
62	Marcilla et al. (2007)	4	IZTECH
63	Mastral et al.-2006 (Gas)	1	IZTECH
64	Mertinkat et al. (1999)	3	IZTECH
65	Miskolczi et al. (2009)	2	IZTECH
66	Miskolczi et al. (2016)	1	IZTECH
67	Murata et al. (2010)	3	IZTECH
68	Serrano et al. (2001)	8	IZTECH
69	Takuma et al. (2000)	1	IZTECH
70	Uemichi et al. (1984)	1	IZTECH
71	Zaiter (2014)	1	IZTECH

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Table B.1 continued

ID	Reference	<i>n</i>	Source
72	Conesa et al. (1997)	4	IZTECH
73	Westerhout et al. (1998)	4	IZTECH
74	Jeong et al. (2022)	7	IZTECH
75	Kaminsky (1985)	4	IZTECH
76	Kaminsky (1991)	11	IZTECH
77	Kaminsky (1995)	3	IZTECH
79	Miller et al. (2005)	3	IZTECH
80	Hardman and Wilson (1998)	1	IZTECH
81	Scott et al. (1990)	9	IZTECH
82	Bockhorn et al. (1998)	1	IZTECH
83	Sodero et al. (1996)	3	IZTECH
84	Westerhout et al.-1998b	14	IZTECH
85	Zeller et al. (2021)	3	IZTECH
86	Angyal et al. (2007)	4	IZTECH
87	Kaminsky and Kim (1999)	4	IZTECH
88	Hernandez et al. (2006)	5	IZTECH
89	Hirota and Fagan (1992)	1	IZTECH
90	Murata et al.-2009a	4	IZTECH
91	Murata et al.-2009b	1	IZTECH
92	Murata et al. (2002)	13	IZTECH
240	Abdullah et al. (2018)	10	AI-Screen
241	Aisien et al. (2021)	2	AI-Screen
242	Akgun et al. (2021)	9	AI-Screen
243	Albor et al. (2023)	9	AI-Screen
245	Ates et al. (2013)	6	AI-Screen
246	Chaiya et al. (2020)	3	AI-Screen
247	Choi et al. (2024)	3	AI-Screen
249	Costa et al. (2007)	4	AI-Screen
250	Cozzani et al. (1997)	9	AI-Screen
251	Dash et al. (2015)	4	AI-Screen
252	de Freitas Costa et al. (2023)	3	AI-Screen
253	Dutta et al. (2021)	13	AI-Screen
254	Horvat et al. (1999)	6	AI-Screen
255	Kamo et al. (2004)	3	AI-Screen
256	Karaduman et al. (2001)	10	AI-Screen
257	Li et al. (2021)	5	AI-Screen
258	Liu et al. (2022)	3	AI-Screen
259	Lodh et al. (2023)	4	AI-Screen
260	Lopez-Urionabarrenechea et al. (2011)	3	AI-Screen
261	Madhu et al. (2022)	7	AI-Screen
262	Maniscalco et al. (2021)	3	AI-Screen

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Table B.1 continued

ID	Reference	<i>n</i>	Source
263	Marais et al. (2024)	6	AI-Screen
264	Matthiesen et al. (2024)	7	AI-Screen
265	Mousavi et al. (2022)	15	AI-Screen
266	Ortega et al. (2023)	6	AI-Screen
267	Papuga et al. (2016)	5	AI-Screen
268	Papuga et al. (2024)	16	AI-Screen
269	Parku et al. (2025)	9	AI-Screen
270	Rangarajan et al. (1998)	3	AI-Screen
272	Shi et al. (2023)	9	AI-Screen
273	Thahir et al. (2019)	7	AI-Screen
275	Verma et al. (2024)	7	AI-Screen
276	Walendziewski et al. (2001)	8	AI-Screen
277	Wang et al. (2023)	8	AI-Screen
278	Zangi et al. (2025)	8	AI-Screen
279	Zayoud et al. (2022)	7	AI-Screen